Home Network Architecture & Devices Assignment

Comprehensive Long-Form Technical Report & In-Depth Solutions

Task 1: Home Network Architecture Diagram & Device Connectivity

A modern home network integrates various hardware components to bridge local client devices with the external public internet via an Internet Service Provider (ISP). Below is a structural breakdown of a typical home network layout comprising a broadband router, an Ethernet network switch, and three end-user client devices.

Detailed Device Connectivity Analysis:

- **1. Smart TV (Wired Ethernet Connection):** The Smart TV connects directly to the Ethernet switch using a physical Category 6 (Cat6) copper cable plugged into an RJ-45 port, ensuring ultra-stable, high-bandwidth streaming without wireless packet drops.
- **2. Laptop (Wired / Wireless Hybrid):** The laptop can connect via a high-speed Cat6 Ethernet cable routed through the network switch for intensive tasks (like file downloads and coding), or connect wirelessly via Wi-Fi 5/6 to the router's access point.
- **3. Smartphone (Wireless Wi-Fi Connection):** The smartphone connects wirelessly via Wi-Fi radio frequencies ($2.4 ext{ GHz}$ or $5 ext{ GHz}$ bands) directly to the router's built-in wireless access point, enabling mobile roaming throughout the home.

Task 2: Comparative Analysis of Core Network Devices & Real-World Use Cases

Different hardware devices operate at varying layers of the OSI model to manage, direct, and expand network traffic. The comparative table below outlines four primary network devices along with their daily application use cases:

Device Name	OSI Layer	Primary Core Function	Real-World Daily Use Case & Service Example
Router	Layer 3 (Network)	Connects distinct networks, routes packets via IP addresses, and provides NAT firewall security.	Connecting a home LAN to the public internet to watch live streaming videos on **YouTube** or browse web content.
Switch	Layer 2 (Data Link)	Connects devices within a LAN using MAC addresses and forwards frames only to the target port.	Connecting multiple desktop PCs and consoles in a local gaming room for low-latency **LAN multiplayer gaming**.

Device Name	OSI Layer	Primary Core Function	Real-World Daily Use Case & Service Example
Hub	Layer 1 (Physical)	Broadcasts incoming electrical signals across all active ports without traffic filtering (obsolete).	Legacy classroom setups where multiple student terminals shared basic broadcast signals in a shared collision domain.
Access Point (AP)	Layer 2 (Data Link)	Creates a wireless local area network (WLAN) allowing Wi-Fi devices to join a wired network.	Providing seamless wireless internet coverage across a multi-story home or college campus for browsing on a **smartphone**.

Task 3: Data Journey Scenario: Ordering Food on Zomato

Scenario: You open the **Zomato** app on your smartphone while connected to home Wi-Fi, search for restaurants, and place a food order. Below is the step-by-step journey of data packets and the specific roles played by each network device involved:

- **1. Smartphone & Wireless Transmission:** When you tap 'Place Order', your smartphone packages your request into digital packets and transmits them via Wi-Fi radio waves ($ ext{IEEE } 802.11$) through the air.
- **2. The Role of the Access Point (AP):** The wireless access point (built into your home router) captures these radio frequency signals using its antenna, demodulates them, and converts the wireless radio signals into standard digital Ethernet frames.
- **3. The Role of the Network Switch:** If your home network utilizes an external Ethernet switch connecting multiple wired and wireless gateways, the switch examines the MAC address of the Ethernet frame and forwards it directly to the router's local port without unnecessary broadcasting.
- **4. The Role of the Router:** The router receives the data packet, reads the destination IP address of Zomato's cloud servers, performs Network Address Translation (NAT) to replace your local private IP with your public ISP IP address, and routes the packet across the internet backbone.
- **5. Response Journey Back:** Zomato's server processes your food order and sends an acknowledgment packet back through the ISP, entering your router, passing through the switch/AP, and arriving back at your smartphone screen confirming your order.

Task 4: Technical Solutions for Connecting a Distant Ethernet Gaming PC

Scenario: You have a standard Wi-Fi router, but your high-performance gaming PC is located in a distant room and only possesses an Ethernet port (no built-in Wi-Fi card). Below are two distinct networking devices you can deploy to solve this connectivity challenge:

Solution 1: Wi-Fi Extender / Repeater with Ethernet Client Port

- **How it works:** A Wi-Fi extender is plugged into a wall outlet halfway between your router and the gaming room. It connects wirelessly to your main Wi-Fi router, capturing the wireless signal.
- **Connecting the PC:** The extender features an integrated RJ-45 Ethernet port on its underside. You plug a short Ethernet patch cable from your gaming PC's LAN port directly into the extender's Ethernet port.
- **Technical Benefit:** This allows your Ethernet-only gaming PC to bridge onto the wireless network without requiring long physical cables running across the house.

Solution 2: Powerline Ethernet Adapter Kit (Recommended for Gaming)

- **How it works:** A powerline adapter kit consists of two wall-plug adapters. Adapter A is plugged into a wall electrical outlet near your Wi-Fi router and connected to the router via an Ethernet cable. Adapter B is plugged into an electrical outlet in the distant gaming room and connected to your gaming PC via an Ethernet cable.
- **Signal Transmission:** The adapters encode digital Ethernet data packets into high-frequency carrier signals and transmit them across your home's existing AC electrical wiring infrastructure.
- **Technical Benefit:** Unlike Wi-Fi extenders which suffer from wireless interference and latency spikes, powerline adapters utilize copper electrical wires, providing a much more stable, low-latency wired connection ideal for competitive online gaming.